CityUniversity of Seattle

School of Technology and Computing

AI Skills Every IT Professional Needs by 2027

A research-backed guide to the capabilities reshaping IT careers — and how to build them

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Why This Guide Exists

Generative AI has moved from experiment to infrastructure inside a few years, and it is rewriting what "competent IT professional" means. This isn't a distant forecast — employers, analysts, and workers are already reporting the shift in hiring data, salaries, and day-to-day workflows.

> *Gartner projects that 80% of the engineering workforce will need to upskill in generative AI-related technologies through 2027 to remain effective.*
>
> *The World Economic Forum's 2026 analysis finds candidates with AI-related skills command advertised salaries 23% higher, on average, than otherwise comparable candidates without them.*
>
> *LinkedIn's Skills on the Rise 2026 report places AI literacy at the very top of the fastest-growing skills list, with job postings requiring it up more than 70% year over year.*

What makes this moment different from past technology shifts is speed. Cloud computing took the better part of a decade to become a baseline IT expectation. Generative AI is compressing that timeline into a few hiring cycles — which means professionals who wait for a slow, gradual on-ramp are more likely to find the on-ramp has already closed by the time they look for it.

The message across these sources is consistent: AI fluency is becoming a baseline expectation for IT roles, not a specialty reserved for data scientists. This guide breaks down seven skill areas the research points to most consistently — AI literacy, AI engineering, retrieval-augmented generation, security, cloud integration, governance, and certification — and maps each one to a path for building it.

1. AI Literacy and Prompt Engineering

AI literacy — the ability to use AI tools effectively, judge their output critically, and understand their limits — now tops LinkedIn's list of fastest-growing workplace skills. For IT professionals, this means going beyond casual chatbot use into structured prompt design, tool orchestration, and knowing when AI-generated output needs a human check.

> *Demand for prompt engineering, chatbot development, and large language model (LLM) skills is rising sharply across IT job postings, according to LinkedIn's 2026 data.*

What this looks like in practice

At a technical level, prompt engineering has moved well past typing a question into a chat window. It now includes techniques like few-shot prompting (showing the model examples of the output you want), chain-of-thought prompting (asking the model to reason step by step before answering), and system-prompt design (setting persistent rules and context for an AI tool used across a team). IT professionals are also increasingly expected to chain multiple prompts and tools together — for example, having one model draft code, a second review it for security issues, and a human make the final call.

- Writing structured, reusable prompts rather than one-off questions
- Evaluating AI output for hallucination, bias, and factual accuracy before it ships
- Understanding context windows, token limits, and when a task exceeds them

- Using tools such as GitHub Copilot, Microsoft Copilot, ChatGPT Enterprise, or Claude to accelerate documentation, code review, and troubleshooting

The common thread across all of it: speed from the tool, judgment from the person. Employers are testing for the second half of that equation as much as the first.

2. AI Engineering and MLOps

A new hybrid role — the “AI engineer” — is emerging at the intersection of software engineering, data science, and machine learning. Gartner specifically calls out this combination as a defining hire for organizations building AI into production systems, distinct from both the traditional software developer and the research-focused data scientist.

> *MLOps specialists handle data validation, model versioning, deployment, and monitoring using tools like scikit-learn, PyTorch, MLflow, and Docker — skills increasingly expected of senior IT and software staff, not just ML researchers.*

The MLOps lifecycle IT professionals are being asked to support

Where DevOps built a discipline around shipping and monitoring software reliably, MLOps applies that same discipline to models — because a model that worked in testing can quietly degrade in production as real-world data drifts away from its training data. That lifecycle typically includes:

- Data validation and preparation — ensuring input data is clean, representative, and free of leakage before training
- Model versioning — tracking which model version is live, and rolling back quickly when a new version underperforms
- Containerized deployment — packaging models with Docker and orchestrating them with Kubernetes so they scale reliably
- Continuous monitoring — watching for model drift, latency spikes, and degraded accuracy after deployment, not just at launch

For working IT professionals, this doesn't mean becoming a research scientist. It means learning how models get deployed, monitored, and maintained in production — the same discipline DevOps brought to software a decade ago, now applied to AI.

3. Retrieval-Augmented Generation (RAG) and Applied AI Development

Gartner identifies retrieval-augmented generation — a technique for grounding AI outputs in an organization's own data — as “essential upskilling” for the engineering workforce through 2027. RAG is what separates a generic chatbot from an AI system that can answer accurately from a company's internal documents, code, or knowledge base, and it is quickly becoming the default architecture for enterprise AI applications.

> *RAG skills are named directly in Gartner's guidance on generative AI upskilling priorities through 2027, alongside broader applied AI development capability.*

> *The vector database market — the infrastructure layer RAG depends on — reached an estimated $3.2 billion in 2025 and continues to grow at roughly 24% annually, as tools like Pinecone, Weaviate, and pgvector move from experimental to production-grade.*

The building blocks of a RAG pipeline

- Embeddings — converting text, code, or documents into numerical representations a model can search against
- Chunking strategy — breaking large documents into pieces small enough to retrieve precisely, without losing context
- Vector database selection — choosing infrastructure like Pinecone (managed, high-scale), Weaviate (strong hybrid search), or pgvector (simplest, Postgres-native) based on team size and workload
- Grounding and evaluation — testing whether the system's answers are actually supported by retrieved documents, not just plausible-sounding

IT professionals who can build or maintain RAG pipelines sit at the center of most enterprise AI projects underway right now, because it's the piece that turns a generic AI tool into one that knows a specific business.

4. AI-Aware Cybersecurity

AI is a double-edged sword in security: it's both a new attack surface and a new defensive tool. Meanwhile, the underlying cybersecurity talent shortage hasn't gone away — it has simply gained a new, urgent dimension.

> *ISC2's 2024 Cybersecurity Workforce Study puts the global workforce at roughly 5.5 million professionals against a gap of 4.8 million unfilled roles — a 19% year-over-year increase in the shortfall.*
>
> *The OWASP Top 10 for LLM Applications names prompt injection as the single most actively exploited vulnerability in AI systems today — occurring when an attacker manipulates an AI tool's input, directly or through a poisoned document or webpage, to override its intended instructions.*

What AI-aware security work covers

The OWASP list extends well beyond prompt injection to include sensitive information disclosure, AI supply-chain vulnerabilities, data and model poisoning, excessive agent permissions, and unbounded resource consumption. None of these are hypothetical — they map directly onto how organizations are already deploying AI copilots and autonomous agents.

- Securing RAG and LLM pipelines against prompt injection and data leakage
- Auditing AI agent permissions so a compromised assistant can't take unauthorized actions
- Running adversarial testing against AI features before they reach production
- Applying defense-in-depth: least-privilege tooling, output filtering, and human approval gates for high-risk AI actions

Security professionals who add AI-specific knowledge to a traditional cybersecurity foundation are positioned at the overlap of two skills gaps at once — the long-standing cybersecurity shortage and the emerging AI-security specialty.

5. Cloud and AI Integration

AI workloads run on cloud infrastructure, and most enterprise AI adoption is happening as an extension of existing cloud platforms rather than a separate initiative. That's also where the costs — and the skills gap around managing them — are landing hardest.

> *Industry workforce analyses point to more than 75% of organizations planning to scale cloud, AI, and cybersecurity capabilities together through 2027, treating them as one integrated skill set rather than three separate tracks.*
>
> *The FinOps Foundation's 2026 State of FinOps report finds that 98% of organizations now actively manage AI spend, up from just 31% two years earlier — and names AI cost management as the single largest skills gap FinOps practitioners are trying to close.*

Why AI workloads break traditional cloud-cost habits

A single high-end GPU can run $2–3 an hour in the cloud; a modest training cluster can burn through several thousand dollars a day; and an inference workload serving real users can cost tens of thousands of dollars a month before anyone notices it's misconfigured. IT professionals who understand both the technical and financial side of AI infrastructure are unusually valuable right now.

- Working with platform-native AI services — AWS Bedrock, Azure OpenAI Service, and Google Cloud Vertex AI
- Understanding GPU provisioning, autoscaling, and when a workload actually needs one
- Cost allocation and chargeback — tagging AI workloads so spend is visible by team or project, not buried in a single cloud bill
- Designing hybrid architectures that mix cloud AI services with on-premises or self-hosted models where it's more cost-effective

6. AI Governance, Ethics, and Critical Thinking

As AI takes over more routine cognitive work, the ability to think independently of it becomes a distinct, marketable skill — not a soft add-on. At the same time, AI governance is shifting from “nice to have” to a legal and contractual requirement in much of the world.

> *The EU AI Act's obligations for high-risk AI systems become fully enforceable on August 2, 2026, making it binding law rather than guidance — with additional obligations following in 2027.*
>
> *The NIST AI Risk Management Framework remains voluntary in the U.S., but is increasingly treated as the de facto standard that federal agencies, procurement teams, and enterprise customers expect vendors and IT teams to already be following.*
>
> *Gartner projects that concerns over critical-thinking atrophy from heavy generative AI use will push roughly half of global organizations to require “AI-free” skills assessments during hiring.*
>
> *Separately, Gartner forecasts that by 2027, half of enterprises without a deliberate, people-centered AI strategy will lose their top AI talent to competitors that invest in workforce enablement rather than tool rollout alone.*

What this means day to day

- Maintaining an inventory of AI systems in use, classified by risk level under frameworks like the EU AI Act
- Documenting intended use, data sources, and decision scope for each AI system — not just its output

- Auditing for model bias and knowing where human sign-off is legally or ethically non-negotiable
- Tracking emerging credentials like ISO/IEC 42001, which is becoming a governance “passport” multinational employers look for

For IT professionals, this translates into being conversationally fluent in AI governance — not a lawyer, but someone who can flag a compliance gap before it becomes an incident.

7. Certifications and Continuous Learning

Employers are formalizing how they screen for AI capability, which means credentials are starting to carry real weight in hiring decisions — not just implied skill from a resume line.

> *Gartner predicts that by 2027, 75% of hiring processes will include certifications or testing specifically for workplace AI proficiency.*

Credentials worth knowing about

- AWS Certified AI Practitioner (AIF-C01) — a foundational, vendor-specific credential covering AI/ML concepts and AWS's generative AI services
- Microsoft Certified: Azure AI Fundamentals (AI-900) and Azure AI Engineer Associate (AI-102) — the second builds and deploys AI solutions using Azure AI services and Azure OpenAI
- Google Cloud Generative AI Leader and related Vertex AI credentials — for teams building on Google's AI stack
- Vendor-neutral options such as CompTIA AI Essentials, useful for professionals not yet committed to a single cloud platform

The practical takeaway: structured, verifiable learning — degree coursework, certificates, applied projects — will increasingly outweigh informal self-teaching in how candidates get screened and hired. Certifications are a fast, visible signal; a degree program is where the underlying fluency across all seven areas in this guide actually gets built.

How CityU's School of Technology and Computing Builds These Skills

CityU of Seattle's School of Technology and Computing offers online bachelor's, master's, and doctoral programs across the exact disciplines this guide covers — designed for working adults and career changers, not just traditional students.

- Applied Computer Science (BS) — foundational programming, data, and systems skills that underpin AI literacy and applied development.
- Computer Science (MS) — advanced coursework touching AI engineering, machine learning, and software architecture.
- Cybersecurity (BS/MS) — an NSA/DHS-designated CAE-CDE program, positioned directly at the AI-security overlap described above.
- Information Technology / Information Systems — the cloud, infrastructure, and governance skills that anchor enterprise AI adoption.
- Artificial Intelligence and Data Science tracks — focused study in the technical core of modern AI systems.

- Graduate certificates — shorter, credential-focused options for professionals who want to move faster than a full degree, mirroring the certification trend described above.

Because programs are online and many bachelor's students can finish in as little as two years with transfer credit, the path from “read the guide” to “have the credential” is designed to be realistic for people already working.

Your Next Step

The skills gap described in this guide is well documented — the harder part is closing it. If you're weighing where AI fits into your IT career, or which program lines up with the direction you want to go, CityU's admissions team can walk through options in a single conversation.

Request more information to get a personalized breakdown of programs, timelines, and transfer credit — no obligation.

Sources

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